MIL-L-13188C(MR)
17 July 1984
SUPERSEDING
MIL-L-13188B(MR)
31 March 1969

MILITARY SPECIFICATION

LEATHER, CATTLEHIDE, CHROME-TANNED: HYDRAULIC PACKING (FOR RECOIL MECHANISMS)

This specification is approved for use by Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers chrome-tanned, cattlehide leather for use as hydraulic packing in recoil mechanisms of ordnance material.

1.2 Classification. Cattlehide leather covered by this specification shall be of the following types and classes as specified (see 6.2):

Type I - Selected leather pieces.
  Class 1 - Nonimpregnated
Type II - Formed leather parts.
  Class 1 - Nonimpregnated
  Class 2 - Impregnated

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified(see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

/FSC 5330/

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SPECIFICATIONS

FEDERAL

T-T-911	-	Twine, Fibrous Jute
UU-P-268	-	Paper, Kraft, Wrapping
PPP-B-585	-	Box, Wood, Wirebound
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Box, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1055	-	Barrier Material, Waterproof, Flexable
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-T-76	-	Tape Packaging Paper (For Carton Sealing)
PPP-T-97	-	Tape, Pressure Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missle and Ordnance
MIL-L-10547	-	Liner, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-S-13282	-	Silver and Silver Alloys

STANDARDS

FEDERAL

Fed. Test Method Std. No. 311 - Leather, Methods of Sampling and Testing

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

American Society for Testing and Materials (ASTM) Standards

ASTM A 109 - Steel, Carbon, Cold-Rolled Strip

ASTM D 3951 - Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103

(Technical society and technical association specifications and standards are generally available for reference from libraries. They also are distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order (see 6.2) and before production has commenced, a sample or samples of the specified item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. The approval of the preproduction samples authorizes the commencement of production but, does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification.

3.2 Material. The leather pieces and formed parts shall be obtained from selected pieces of belting butt bends made from cattlehides. The leather shall not be buffed, and shall have no cuts, brand scars, grub holes, open scratches, wrinkles, or spongy spots. The belting butt bends shall conform to the definitions given in Fed. Test Method Std. No. 311.

3.3 Moldability (Type 1). Type 1 leather shall be capable of being molded in a mold. The type of mold to be cited by the procuring activity (see 6.4). Test shall be made as specified in 4.8.1.

3.4 Size and thickness.

3.4.1 Type 1. The leather pieces shall be furnished in the size and thickness specified (see 6.2). The minimum size to be specified shall be 6.75 inches (17.15 cm). The thickness shall be as specified by the procuring activity.

3.4.2 Type II. The formed leather parts shall be of the dimensions and form, as specified (see 6.2).

3.5 Tannage. The leather shall be tanned with chromium salts, without the use of vegetable tanning materials.

3.6 Coloring. The coloring of the leather, if any, shall be applied during the tanning process. The coloring shall be insoluble in water, oil, chloroform, and acetone.

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3.7 Finish. The leather shall be full-grain. The flesh side shall have no flesh, and may be split, shaved, or skived to secure greater uniformity of thickness.

3.7.1 Hard spots or grit. The leather shall have no hard spots or grit which shall scratch highly polished metal surfaces when tested as specified in 4.8.2.

3.8 Impregnation of leather (Class 2 only). Class 2 leather shall be impregnated so as to comply with the inspection as specified in 4.7.2 (see 6.3).

3.9 Cracking.

3.9.1 As furnished. The leather shall not crack on either the grain, or the flesh side when folded back on itself, and shall show no wrinkles when flattened out after being so bent (see 4.8.3.1).

3.9.2 After heating at 150°C (302°F). When tested as specified in 4.8.3.2 the heated leather shall not crack on either the grain or the flesh side.

3.10 Shrinkage. The shrinkage of the leather shall not exceed 5.0 percent, by area, when tested as specified in 4.8.4.

3.11 Tensile strength. The average tensile strength shall be not less than 2.5 ksi (0.172 MPa), when tested as specified in 4.8.5.

3.12 Chloroform extractable material (grease) Class 1 only (Nonimpregnated leather). The chloroform extractable material (grease) content of the nonimpregnated leather on a moisture free basis shall be within the range of 3 to 10 percent when tested as specified in 4.8.5. The grease (oil or fat liquor) should not interfere with the shop impregnation or other procedures (see 4.3). The grease content shall also be so that the leather shall pass the moldability test (see 4.8.1) and the cracking test (see 4.8.3).

3.13 Chromic oxide. The chromic oxide content shall be not less than 3.5 percent, calculated on a moisture-and-chloroform extractable-free basis, when tested as specified in 4.8.5.

3.14 Ash. The ash content of the leather, calculated on a moisture-and-chloroform extractable-free basis, shall not exceed the ratio of 1.2 parts ash to 1 part chromic oxide (see 3.13), when tested as specified in 4.8.5.

3.15 Chlorides. The soluble chlorides content, expressed as sodium chloride, shall not exceed 0.30 percent, calculated on a moisture-and-chloroform extractable-free basis, when tested as specified in 4.8.5.

3.16 Sulfates. The soluble sulfates content, expressed as sulfuric acid, shall not exceed 0.30 percent, calculated on a moisture-and-chloroform extractable-free basis, when tested as specified in 4.8.5.

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3.17 Acidity. The pH value of the leather shall be not less than 3.3, when tested as specified in 4.8.5.

3.18 Corrosiveness. The leather shall show no pitting, etching or staining to silver-copper alloy and cold rolled steel when tested as specified in 4.8.6.

3.19 Workmanship. The workmanship shall be in accordance with high-grade commercial practices to produce leather pieces and formed parts which will meet the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections.

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First Article Inspection.

4.3.1 First Article Tests: First Article Tests (see 6.5) shall consist of all physical and chemical tests of this specification.

4.4 Quality Conformance Inspection.

4.4.1 Quality Conformance Inspection shall consist of inspection and tests for all requirements of this specification.

4.5 Lot.

4.5.1 Lot. A lot shall consist of all leather units of the same type and thickness, manufactured from a single batch of hides and tanning and finishing materials, and from a single production method. The lot shall be offered for acceptance at one time and place.

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4.6 Sampling.

4.6.1 Sampling for visual and dimensional inspection. Samples for visual and dimensional inspection shall be selected at random in accordance with Fed. Test Method Std. No. 311, and of sample size as specified in table I of Fed. Test Method Std. No. 311. The unit of product shall be one piece or part.

4.6.2 Sampling for physical and chemical tests. Unless otherwise specified samples for physical and chemical tests shall be selected at random in accordance with Fed. Test Method Std. No. 311 and of the sample size as specified in Table II of Fed. Test Method Std. No. 311. The unit of product shall be one piece or part.

4.7 Inspection.

4.7.1 Visual and dimensional inspection. Samples shall be selected in accordance with 4.6.1. The acceptable quality level shall be as specified in Table I of Fed. Test Method Std. No. 311. Examination shall be for the defects listed in Table I of this specification.

Table I. Classification of Defects

Category	Defect	Reference Paragraph	Method of Inspection
Critical	None defined		
Major 101	Leather not from selected pieces of belting butt bends	3.2	Visual
Major 102	Leather buffed	3.2	Visual
Major 103	Cuts, brand scars, grup holes, or open scratches	3.2	Visual
Major 104	Wrinkles or spongy spots	3.2	Visual
Major 105	(Type I) Wrong size or thickness	3.4.1	Measure
Major 106	(Type II) Wrong dimensions or form	3.4.2	Measure
Major 107	Leather not full-grain	3.7	Visual
Major 108	Leather showing flesh	3.7	Visual
Major 109	(Class 1) contains impregnating material	1.2	Visual
Major 110	(Class 2) incomplete impregnation	4.7.2	Visual
Major 111	Poor workmanship	3.19	Visual
Minor	None defined		Visual

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4.7.2 Impregnated leather (class 2). The presence of impregnant throughout the entire thickness of leather (saturation) shall be verified by examining a cut cross-section of a trial piece of leather. Impregnated leather will appear much darker in color than unimpregnated leather. If the entire cross-section has darkened, impregnation has been completed (see 6.3).

4.8 Tests.

4.8.1 Moldability (type I only). The leather pieces submitted as bid samples (see 6.5) shall be subjected to a working test (see 3.3) for determining moldability by standard shop practices, as directed by the procuring activity.

4.8.2 Hard spots or grit. Steel plates 3 inches (7.62 cm) long by 2 inches (5.06 cm) wide conforming to cold rolled 1020 of ASTM A 109 with a microinch finish of 2 to 4 shall be rubbed parallel to the long direction against the leather specimens. Butt bend specimens 2.7 to 3.0 inches (6.86 to 7.62 cm) in diameter shall be used. The rubbed surface of the plates shall be visually examined for presence of scratches. Note: if partly masked with pressure sensitive tape during this test, the plates may be unmasked for use in the test for corrosiveness (4.8.6.).

4.8.3 Cracking.

4.8.3.1 As furnished. Test the leather according to method 4021 of Fed Test Method Std. No. 311. The diameter of the mandrel shall be 0.0625 inch (0.159 cm) larger than the specimen thickness. Check for wrinkles after bending separate specimens both with the grain out and with the flesh side out. Note: The test should be performed parallel to and perpendicular to the backbone direction.

4.8.3.2 After heating at 150°C (302°F). The leather test specimen shall be heated for 45 minutes in an oven maintained at 150° ± 3°C (302° ± 5.4°F). The leather shall be heated dry; i.e., without being submerged in oil. immediately after removal from the oven the leather shall be tested for cracking in accordance with method 4011 of Fed. Test Method Std. No. 311. The diameter of the mandrel used in bending the leather shall be 0.0625 inch (0.159 cm) larger than the minimum thickness of the leather specimen.

4.8.4 Shrinkage. The length and width of the test specimen shall be measured to the nearest 0.0625 inch (0.159 cm), along the center lines of the specimen. The specimen shall then be submerged in water and suspended (that is, not touching any part of the container) for 24 hours. The water shall be kept at a temperature of 20° to 30°C (68° to 86°F). The water containing the leather specimen shall then be heated to boiling, increasing its temperature at the rate of 5° to 7°C (9° to 12.6°F) per minute, and then boiled for 6 minutes. The specimen shall then be removed, and its length and width remeasured immediately. The shrinkage shall be calculated as percentage of the original area of the test specimen.

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4.8.5 Other physical and chemical tests. The tests listed in Table II shall be performed in accordance with the corresponding specified method of Fed. Test Method Std. No. 311.

Table II. Other physical and chemical tests

Test	Requirement paragraph	Fed. Test Method Std. No. 311
Tensile strength	3.11	2021
Chloroform extractable material (Grease)	3.12	6341
Chromic oxide content	3.13	6511
Ash content[1/]	3.14	6421
Soluble chlorides content	3.15	6351
Soluble sulfates content	3.16	6361
Acidity (pH)	3.17	6621

1/ Impregnated-leather test specimens shall be degreased and dewaxed prior to testing for ash content.

4.8.6 Corrosiveness. Make two disks of the leather under test, each disk being approximately 0.625 inch (1.588 cm) in diameter. Place the disks between a polished plate of silver-copper alloy (90-percent silver, 10-percent copper) conforming to MIL-S-13282, grade C in the annealed temper and a polished steel plate, made of cold rolled 1020 steel of ASTM A 109 with a microinch finish of 2 to 4, (See note of 4.8.2) so that each plate is in contact with the grain side of one disk and the flesh side of the other. Lightly press the metal plates against the leather disks by means of a small metal clamp, thus establishing a metal connection between the two plates, and insuring good contact between the leather and the plates. Submerge the assembly for 72 hours in hydraulic oil conforming to MIL-H-5606, held at a temperature of 100° ± 3°C (212° ± 5°F). Following the exposure period the metal plates shall be examined for evidence of corrosion, such as etching, pitting, or staining. If any corrosion exists, the surfaces of the plates shall be washed in precipitation naptha. If only a slight discoloration of the plates remains after this treatment, the leather shall be considered to have met the requirements of 3.18. Any marks remaining on the surface of the plates after the naptha wash shall be lightly polished with a nonabrasive cloth buff. Any pits or eroded marks remaining after the naptha wash shall be construed to be corrosion. Discoloration or staining marks which do not physically affect the surface of the plates and which easily wash or buff off shall not be considered detrimental. If any doubt should arise about the presence of pitting, erosion or corrosion on the metal plates, a microscope of approximately 10 to 15 power magnification shall be used to determine the actual condition.

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4.9 Examination of preparation for delivery. The packaging, packing, and marking of packing material shall be examined to determine compliance with the requirements of Section 5 of this specification.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I (Leather pieces). Leather pieces of approximately equal size and thickness shall be bundled together. Each bundle shall be over-wrapped with a 60-pound kraft paper conforming to grade A of UU-P-268. The wrap shall be sealed with pressure sensitive tape conforming to PPP-T-76, and tied with 4-ply twine conforming to type C of T-T-911. The twine shall be applied double in at least two directions around the bundle, and knotted on the sides and ends of the bundle. The gross weight of the bundle shall not exceed 12 pounds (5.44 kg).

5.1.1.2 Type II (Formed leather parts). The formed leather parts shall be packaged in dust and moisture-free metal containers conforming to any suitable type of PPP-C-96. It is essential that the containers be capable of holding the leather parts in their formed shape from the time they are formed and packaged until they are removed for assembly in the corresponding recoil mechanism. The gross weight of the metal container shall not exceed 12 pounds (5.44 kg).

5.1.2 Commercial (type I and type II). When specified, packaging shall be sufficient to afford physical protection against damage during shipment from the supply source to the first receiving activity. This level to conform to ASTM D 3951 when such meets the requirements of this level (see 6.2).

5.2 Packing. Packing shall be level A, B or in accordance with ASTM D industrial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I - Leather pieces. Leather pieces packaged as specified in 5.1 shall be packed in any of the following overseas type boxes; wood wirebound; wood cleated fiberboard; wood cleated plywood; wood, nailed and lock corner; fiberboard; conforming to the requirements of PPP-B-585, style 2, 2A, or 3, PPP-B-591 class II, weather-resistant, PPP-B-601, PPP-B-621, style 4 or PPP-B-636, class II, weather-resistant, respectively. Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall conform to type E-2, H-3 or L-2 of PPP-B-1055 and be closed and sealed in accordance with the appendix of MIL-L-10547. Caseliners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1.5 inch (3.81cm) wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiberboard boxes shall be banded with tape conforming to type III of PPP-T-97 and the appendix thereto. The gross weight of each shipping container shall not exceed 70 pounds (31.75 kg).

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5.2.1.2 Type II (Formed leather parts). The formed leather parts packaged as specified in 5.1 shall be packed in any of the following overseas type boxes; wood wirebound; wood cleated fiberboard; wood cleated plywood, wood, nailed and lock corner; or fiberboard conforming to the requirements of PPP-B-585, style 2, 2A or 3, PPP-B-591, PPP-B-601, PPP-B-621, style 4, of PPP-B-636, class weather-resistant, respectively. No caseliners shall be furnished. Adequate blocking or cushioning of the interior containers shall be provided to prevent shifting of the load during transit. The gross weight of each shipping container shall not exceed 70 pounds (31.75 kg).

5.2.2 Level B.

5.2.2.1 Type I or type II. Leather pieces or formed parts packaged as specified in 5.1, as applicable, shall be packed in domestic type boxes conforming to the following PPP-B-585, style 2, 2A or 3, class 2, PPP-B-591, PPP-B-601, PPP-B-621, style 4 or PPP-B-636, class domestic. The gross weight shall not exceed 150 pounds (68.04 kg).

5.2.3 Industrial.

5.2.3.1 Type I or Type II. Leather pieces or formed parts packaged as specified in 5.1, as applicable, shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Marking. Shipping containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use: This specification covers leather pieces and formed parts intended for use in hydraulic recoil mechanisms for Ordnance material.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Type and class of items required (see 1.2).
(c) First Article: Whether First Article is required (see 3.1 and 6.5).
(d) Size and thickness of leather pieces (type I) and dimensions and form of the leather parts (type II) (see 3.4.1 and 3.4.2).
(e) Level of packaging and level of packing required (see 5.1 and 5.2).

6.3 Impregnant. For class 2 leather a means of impregnation that has been found to be satisfactory is to completely submerge the leather in a hot impregnant consisting of a 1:1 by weight mixture of melted carnauba wax and lead oleate, or ouricuri wax and lead oleate for a period of time sufficient to assure complete saturation. The impregnant should be in a melted condition at all times during the impregnation procedure and should be kept at a temperature as low as possible. NOTE: The time required to saturate 0.125 inch (0.318 cm) thick leather is approximately thirty (30) minutes (see 4.7.2).

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6.4 Moldability (Wiper-Drawing No. A5015746). In the moldability test, the leather pieces should conform to the current "shop practices" an example of which is a "wiper" shown in Figure 1, reproduced from Rock Island Arsenal drawing No. A5015746. Typically the part is molded and then machined in a lathe to the drawing dimensions. A simple mold consisting of matching pieces held together by a single 0.625 inch (1.588 cm) bolt and washers is used to form the wiper. For the part shown, leather (conditioned at ambient temperature and humidity) is split to 0.125 inch (0.318 cm) thickness and cut in the form of a blank, 3.25 inches (8.255 cm) in diameter with a 0.625 inch (1.588 cm) center hole. The blank is bolted into the mold at room temperature, and the assembly is brought to 100°C (212°F) on a hot plate. The assembly is allowed to cool to room temperature before parting. The molded blank is then clamped in a lathe and trimmed to the drawing specifications. The leather should be firm enough to permit clamping and tooling without significant distortion.

6.5 First Article Test. When specified (see 6.2) and as soon as practicable after the award of a contract or order, 15 pieces of type I leather samples shall be separately packaged and forwarded to Commanding General, Hq. U.S. Army Weapons Command, ATTN: AMSWE-RET, Rock Island Arsenal, Rock Island, Illinois 61201, who shall have the responsibility for testing, to determine compliance with the requirements of the specifications and drawings and notifying the contracting officer of approval or disapproval. Samples shall be plainly identified by securely attached durable tags, marked with the llowing information:

(1) Samples for First Article Testing

(2) Contract or order number

(3) Specification number/title, i.e., MIL-L-13188C(MR) Leather; Cattlehide, Chrome-Tanned; Hydraulic Packing (For Recoil Mechanisms)

(4) Mfr's name and part number.

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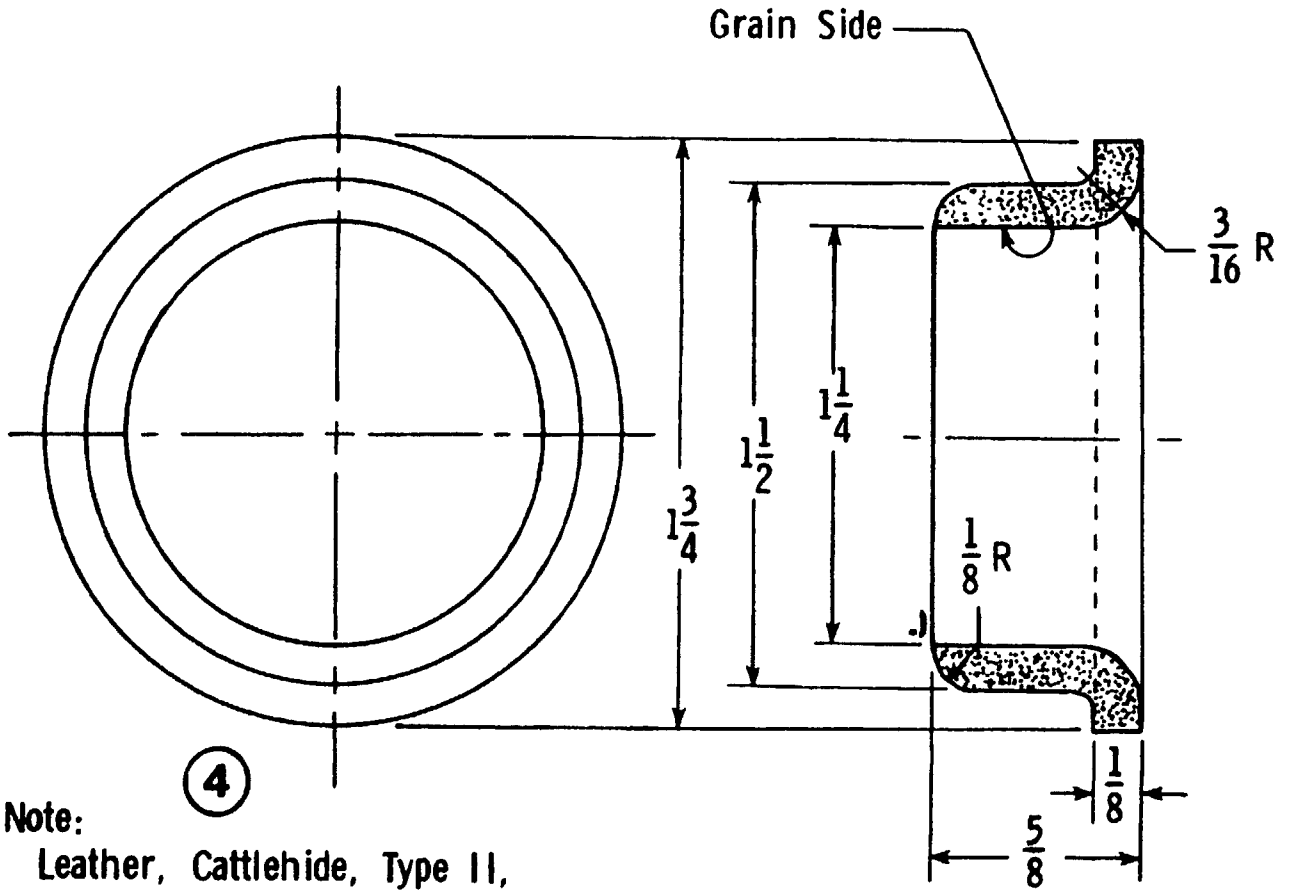
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Note:
Leather, Cattlehide, Type II,
Class 1, Spec MIL-L-13188.

(all dimensions in inches - inches multiplied by 2.54 = centimeters)

Figure 1. PART NO. 5015746

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5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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INCH-POUND

MIL-L-13188C(MR)
AMENDMENT 1
6 February 1990

MILITARY SPECIFICATION

LEATHER, CATTLE HIDE, CHROME-TANNED: HYDRAULIC PACKING
(FOR RECOIL MECHANISMS)

This amendment forms a part of MIL-L-13188C(MR), dated 17 July 1984, and is approved for use by the Materials Technology Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

Paragraph 3.4.2 - delete (see 6.2), and substitute "The thickness shall be a uniform thickness and the faces shall be parallel and flat. The inside and outside diameters shall be perpendicular to the parallel surfaces (see 6.2)."

PAGE 5

Paragraph 3.19 - add the following sentence, "The finished product shall not contain any frayed or rough edges."

PAGE 10

Paragraph 6.3, line 8 - delete "Approximate thirty (30) minutes (see 4.7.2)." and substitute "a minimum of sixty (60) minutes. If the leather does not appear to be fully impregnated, it is returned to the impregnation pot and allowed to soak until it appears to visually meet the accept/reject criteria of impregnation, namely, the entire area of the cross section of the leather is darkened (see 4.7.2)."

Custodian:
Army - MR

Review activities:
Army - AR
DLA - IS

User activity:
Army - AT

Preparing activity:
Army - MR

Project 5330-A100

AMSC N/A

FSC 5330

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